

ICT703 Programming

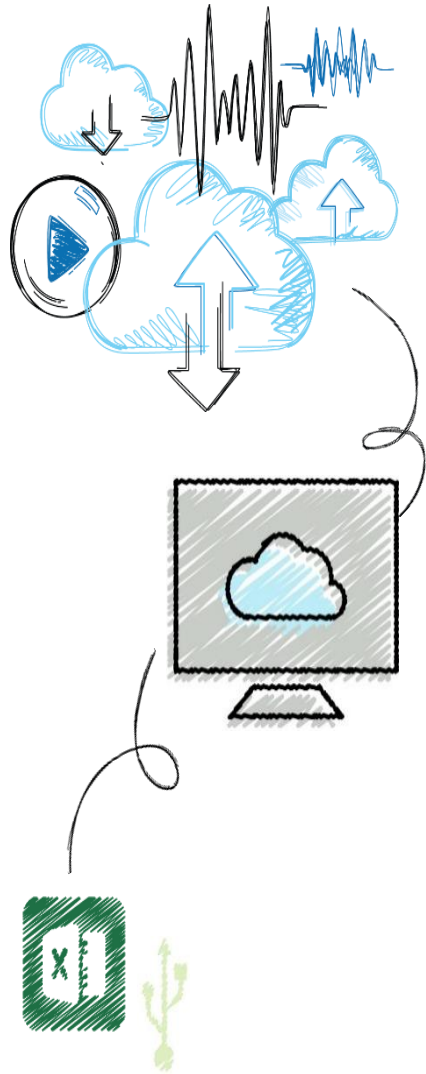
Module 12

Text to Speech



Objectives

- Explore a new module
- Add functionality to an existing program





Text to Speech

- Add audio output to your program.
 - Build it ourselves (we could use our translator program)
 - Use Google Text to Speech – produces an audio file.
 - Online using the Google API
 - **gTTS** stands for Google Text To Speech
 - Use **pyttsx3** - looks for TTS engines pre-installed in your platform
 - Offline
 - Converts and sends audio to the speaker.
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Google Text to Speech - Simple

import the module

```
import gtts
```

make request to google to get synthesis

```
tts = gtts.gTTS("Hello world")
```

save the audio file

```
tts.save("hello.mp3")
```



Google Text to Speech – Play the Sound

import the module

```
import gtts
```

```
from playsound import playsound
```

make request to google to get synthesis

```
tts = gtts.gTTS("Hello world")
```

save the audio file

```
tts.save("hello.mp3")
```

play the audio file

```
playsound("hello.mp3")
```



Google Text to Speech – Different Languages

```
tts = gtts.gTTS("Hello everyone, how are you")
```

```
tts = gtts.gTTS("Herkesse merhaba nasılsınız", lang="tr")
```

```
tts = gtts.gTTS("Hallo zusammen, wie geht es euch",  
lang="de")
```

```
tts = gtts.gTTS("Hallo almal, hoe gaan dit", lang="af")
```

```
tts = gtts.gTTS("Dàjiā hǎo, nǐmen hǎo ma", lang="zh")
```



Text to Speech – using pyttsx3

- Offline – doesn't use an API but use TTS engines on your computer
 - SAPI5 on Windows
 - NSSpeechSynthesizer on Mac
- You can choose among different voices that are installed on your system
- Controlling the speed of speech, Tweaking volume
- Saving the speech audio into a file



Text to Speech – using pyttsx3

import the module

```
import pyttsx3
```

initialize Text-to-speech engine

```
engine = pyttsx3.init()
```

convert this text to speech

```
engine.say("Hello everyone, how are you?")
```

play the speech

```
engine.runAndWait()
```



Text to Speech – Have the option

- An inclusive program would allow the user to choose the output method best suited to them.
- As example, provide text on screen or audio.
- A simple way to do this is to create a global variable that is set to the user's preference and create a function called output that would either print or say.